Gym Management System: Full Plan with AI

Working name: **Ralypto Gym** (change if you want). Sold to gym owners as a monthly subscription.

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1. The Big Idea

Most gym software does the same things: member list, fee collection, check-in. The owner still does all the thinking. **Our system should keep members, make money, and coach people on its own.**

Our promise: "Fewer members leave. More members come. Everyone gets a plan."

- It **spots members who are about to quit** and helps the gym win them back.
- It gives every member a **personal workout and meal plan** (trainer approves it).
- Members **book, ask, and check their plan by chat or voice** in Sinhala, Tamil, and English.
- The owner sees **who is busy, who is late with fees, and what to do next**, in plain words.

What makes it different from competitors: local languages, WhatsApp-first, local food plans, and AI that takes action (drafts messages, books classes), not only reports.

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2. Who Uses It

Role	Main screen	Can do
Owner	Dashboard	Everything, money, reports, plan and billing
Manager	Dashboard	Members, staff, schedule, reports, offers
Front Desk	Check-in	Check-in, new member, take payment, sell items
Trainer	My Clients	Client plans, sessions, progress, notes
Member	Member App (phone)	Check-in QR, bookings, workout plan, progress, pay fees
Super Admin (us)	Admin Panel	Create gyms, manage subscriptions

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3. Core Modules

Priority 1: Must have for the first pilot gym

3.1 Members

- Add member in under 1 minute: name, phone, photo (webcam or upload), goal, emergency contact.
- Health form (simple, PAR-Q style): injuries, conditions, medicines. Member signs on screen (digital waiver).
- Member profile: plan, status (active, expiring, expired, frozen), visits, payments, notes, trainer.
- Search by name, phone, or card number.

3.2 Membership Plans

- Plans: monthly, quarterly, yearly, day pass, student, couple/family, trial.
- Rules: freeze (days), transfer, upgrade, joining fee, discount codes.
- Auto status change: active, expiring soon (7 days), expired, grace period.

3.3 Billing and Payments

- Take payment: cash, card, bank transfer, online link.
- Receipt and invoice (print, WhatsApp, SMS).
- **Due list** (who owes, how much, since when) and automatic reminders.
- Recurring billing (if gateway supports it) or reminder-based renewal.
- Daily cash report and shift close.

3.4 Check-in and Access

- Member shows **QR code** in the app (or card, or phone number). Front desk screen shows a big **green (OK) or red (expired / unpaid)** with photo and name.
- 2 taps maximum. Works **offline**.
- Later: turnstile or door lock integration, fingerprint or face device.
- Live "people in gym now" counter.

3.5 Staff

- Staff list, roles, attendance, shifts, trainer commission, salary sheet.

3.6 Basic Reports

- Active members, new joins, renewals, expired, revenue by day / month, dues, peak hours, plan mix, trainer performance.

Priority 2: Next

3.7 Classes and Bookings

- Weekly timetable (Yoga, Zumba, HIIT...), trainer, room, capacity.
- Member books, cancels, joins waitlist. Cancel rules (late cancel counts as a visit).
- Reminders before class.

3.8 Personal Training

- Sell session packs (10 sessions). Trainer calendar. Book, complete, and track remaining sessions.

3.9 Workout Plans and Logging

- Exercise library (name, muscle, video or image, how-to).
- Plan templates (Fat loss 3 days, Muscle 5 days, Beginner...).
- Assign plan to member. Member logs sets, reps, weight in the app.

3.10 Progress Tracking

- Weight, body measurements, optional photos, attendance streak, personal records.
- Charts and goals ("Lose 5 kg by March").

3.11 Member App (PWA)

- Login by phone and code. QR check-in, my plan, today's workout, book class, pay or renew, progress, messages from gym.
- Install on the phone home screen (no app store at first).

3.12 Leads and Marketing (CRM)

- Lead list (walk-in, Instagram, referral), trial pass, follow-up reminders, status (new, trial, joined, lost).
- WhatsApp and SMS messages and campaigns (offers, renewals, birthdays).
- Referral system ("bring a friend, get a free week").

Priority 3: Later

- **Retail and POS** (protein, drinks, merch) with stock.
- **Lockers** (rent, expiry, assign).

- **Equipment and maintenance** (list, service dates, faults).
- **Multi-branch**.
- **Accounting** (income, expenses, profit).
- **Wearable and device integrations**.

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4. AI Features

Score: Wow = how much it impresses, Effort = how hard (1 easy, 5 hard).

#	Feature	What it does	Wow	Effort
1	**Churn Alert + Win-Back Agent**	Finds members who are likely to quit (visits dropping, plan ending, missed payments). Shows a risk list. AI drafts a friendly WhatsApp message and a suggested offer. Staff taps Send	5	3
2	**AI Workout Coach**	Member answers a short chat (goal, level, days, equipment, injuries). AI builds a plan from the gym's own exercise library. **Trainer reviews and approves** before the member sees it. Plan adapts from logged workouts	5	4
3	**AI Meal Plan (local food)**	Simple daily meal ideas using local foods (rice, dhal, eggs, fish, kottu portions…), budget-aware, vegetarian option. Marked as general guidance, not medical advice	4	3
4	**Member Assistant (chat + voice)**	"What is my workout today?" "Book me Zumba tomorrow 6pm." "Freeze my membership for 2 weeks." Works in Sinhala, Tamil, English, on app and WhatsApp	5	4
5	**Live Crowd + Peak Forecast**	"Gym is quiet now." "Busiest at 6-8pm." Member app shows best times. Owner gets a staffing and class schedule suggestion	4	3

#	Feature	What it does	Wow	Effort
6	**Ask Gym (Owner Assistant)**	"Why did revenue drop this month?" "How many members at risk?" Answers from real data, with small charts. Weekly summary on WhatsApp	5	3
7	**Smart Lead Scoring**	Ranks leads by chance to join and suggests the next follow-up and message	4	2
8	**Smart Dues Reminder**	Learns who pays late, when to remind, and which tone works. Fewer unpaid fees	3	2
9	**Form Check (camera)**	Member films a set; AI counts reps and flags posture mistakes (pose detection). Big wow, but must be careful and honest	5	5
10	**Class Demand and Schedule Advice**	Which classes fill up, which are empty, best time slots, suggest cancel or move	3	3
11	**Off-Peak Offers**	Suggests discounts for quiet hours to move members and earn more	3	2
12	**Motivation Nudges**	Personal messages: streak reminders, milestones, "we miss you", birthday	3	2
13	**Equipment Care Alerts**	Predicts service dates from usage and fault history	2	3
14	**Onboarding Safety Screen**	Reads the health form; flags members who should see a doctor first. Never gives medical advice	4	2
15	**Content Maker**	Drafts Instagram posts, offers, and class announcements in the gym's style	2	1

My Recommendation: 5 Hero Features

Build these first. They make a strong 3-minute demo and they save or earn money for the owner.

1. **Churn Alert + Win-Back Agent** (the money feature)
2. **AI Workout Coach with trainer approval** (the member wow feature)
3. **Member Assistant** (chat first, voice next)
4. **Live Crowd + Peak Forecast**
5. **Ask Gym + Weekly Summary**

Add-on later: Form Check (camera). Great for marketing, but hard and risky. Build it after the heroes work.

What I Would Not Do

- **Do not estimate body fat or health from photos with AI.** It is inaccurate, can hurt people, and creates privacy risk.
- **Do not let AI give medical advice or diet plans for people with health conditions.** Flag and send to a doctor or trainer.
- **Do not let AI change money** (refunds, discounts, price) without a human tap and PIN.

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5. How the AI Works (Simple Version)

5.1 Rules for All AI

1. **AI proposes, human confirms** for anything important (plans, messages, bookings that cost money).
2. **Trainer approves every workout and meal plan** before a member sees it.
3. AI calls the same backend APIs as a normal user, with that user's permissions. It never touches the database directly.
4. AI never invents facts (class times, balances, prices come only from the system).
5. Every AI action is logged with an "AI" tag.
6. Normal tap screens always work if AI or internet is down.
7. Limits and credits per plan, because AI costs money.

5.2 Churn Prediction (Start Simple)

Stage	Method
Start (first 3 months)	**Rules**: visits in last 14 days vs the member's usual, days since last visit, plan ends in under 14 days, late payment, missed booked classes. Give each a score. Show **High / Medium / Low**
Later (needs 3-6 months of data)	**Machine learning** (logistic regression or gradient boosting) trained on members who left vs stayed. Show the **top reasons** for each risk

Be honest in the UI: "Based on visit pattern." Do not call it 100% accurate. Measure it: of members marked High, how many really left?

5.3 AI Coach

```
Member chat answers -> AI + gym's exercise library + member limits (injuries,
equipment)
   -> Draft plan -> Trainer reviews and edits -> Approved plan -> Member app
   -> Logged workouts -> AI suggests next week's changes -> Trainer approves
```

- The AI can only pick exercises **from the gym's library** and must follow safety rules (beginner volume, no listed injury exercises).
- Use structured output (JSON) so the plan fits the app screens.
- Trainers can turn AI coach off per member.

5.4 Member Assistant and Voice

```
Message or voice -> Speech-to-text (if voice) -> AI with tools -> Backend API
-> Reply
```

Tools: `get_my_plan`, `get_today_workout`, `list_classes`, `book_class`, `cancel_booking`, `get_my_dues`, `request_freeze`, `get_gym_hours`. The AI can only act on the **logged-in member's own data**.

Language note: Sinhala and Tamil speech recognition quality differs by provider and changes. Test with 100 real recordings before promising it. Start with text chat in all three languages and English voice, then add Sinhala and Tamil voice when tests pass.

5.5 Crowd and Peak Forecast

- Live count from check-ins and check-outs (or time-based auto check-out after 2-3 hours).
- Forecast = average by weekday and hour, then add holidays, exam seasons, weather, and class times later.

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6. Easy-to-Use Design Rules

Same style as the Ralypto restaurant system: **clean, Apple-like, black and white, simple.**

- **Front desk check-in in 2 taps.** Big search bar. Big green or red result with photo.
- Big buttons (48px or more), icon plus text label, simple words ("Renew", "Check In", "Add Member").
- One main button per screen. Status colours only small: green = OK, amber = expiring, red = expired or unpaid, blue = info.
- **Member app is mobile-first**, with 4-5 bottom tabs: Home, Workout, Classes, Progress, Profile.
- Languages: English, Sinhala, Tamil. Switch anywhere.
- Every screen has loading, empty, and error states.
- Setup wizard: gym info, plans, staff, classes, import members from Excel. Target: 30 minutes.
- Offline check-in at the front desk, with a clear Online / Offline badge.

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7. Subscription Plans for Gym Owners (Draft)

Feature	Basic	Standard	Premium
Members	up to 150	up to 500	Unlimited
Members, plans, billing, check-in	Yes	Yes	Yes
Staff accounts	3	10	Unlimited
Classes and bookings	No	Yes	Yes
Personal training	No	Yes	Yes

Feature	Basic	Standard	Premium
Member app (PWA)	Basic (QR, fees)	Full	Full + branded
Workout logging and progress	No	Yes	Yes
WhatsApp/SMS messages	Pay per message	Included (limit)	Included (higher limit)
Churn alerts (rules)	No	Yes	Yes
Churn ML, Win-Back Agent	No	No	Yes
AI Workout Coach, Meal Plan	No	No	Yes
Member Assistant, Ask Gym	No	Limited	Yes
Crowd and peak forecast	No	Yes	Yes
Multi-branch	No	No	Yes
Support	WhatsApp hours	WhatsApp	Priority

AI costs real money per use. Use **fair-use limits or AI credits**. Free trial 14 days. Yearly discount. Feature locks are done by the backend with feature flags. Show a friendly "Upgrade to unlock" card.

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8. Technical Plan

8.1 Suggested Stack

Part	Suggestion
Staff web app	React + TypeScript, Tailwind, shadcn/ui, TanStack Query, Zustand, framer-motion
Member app	Same codebase as PWA (separate member routes) or a second small PWA
Backend	Friend's choice (NestJS or FastAPI)
Database	PostgreSQL (multi-tenant: `gym_id` on every table)

Part	Suggestion
Cache / queue	Redis (reminders, jobs)
Real-time	WebSocket (live check-ins, crowd count)
AI service	Python + FastAPI, separate folder
Messaging	WhatsApp Business API and an SMS gateway
Payments	Local payment gateway (for example PayHere), bank-transfer marking
Files	S3-compatible storage (member photos, exercise media)
Hosting	Docker on a VPS to start

8.2 Architecture

```
Staff app + Member app (PWA)
        |
Backend API  <---- Tools API with permissions ----  AI service
        |                                          |-- chat and voice
  PostgreSQL + Redis                               |-- churn and forecast
jobs
        |
  WhatsApp / SMS / Payment gateway / (later) door and turnstile devices
```

8.3 Main Data (Backend Starting Point)

Gym, Branch, Plan (SaaS), Subscription, Invoice, User, Role, Staff, Member, MembershipPlan, Membership, Freeze, Payment, Due, Receipt, CheckIn, Visit, HealthForm, Waiver, Class, ClassSession, Booking, Waitlist, TrainerSession, SessionPack, Exercise, WorkoutPlan, WorkoutLog, Measurement, Goal, Lead, Campaign, Message, Locker, Equipment, Product, StockMovement, AIAction (audit), ChurnScore, Notification.

8.4 Data to Keep From Day 1 (for AI)

- Every check-in with a time. Every booking, cancel, and no-show.
- Every payment and late payment.
- Never hard-delete members or visits. Use status flags.

- Reason when a member leaves (a short dropdown at the front desk). This is gold for churn learning.
- `source` on messages and bookings (`staff` , `member_app` , `whatsapp` , `voice` , `ai`).

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9. Safety, Privacy, and Trust

- **Health information is sensitive.** Store only what is needed, restrict who can see it (trainer and manager), and get clear consent.
- Digital waiver with date and signature. Keep a copy.
- **No medical advice.** AI shows a clear note: "General fitness guidance. Talk to a doctor if you have health problems." Members with flagged health answers go to a trainer.
- Members under 18: need parent or guardian consent. Check local rules.
- Photos and progress pictures are private to the member and trainer. Easy delete.
- Do not store card numbers. Use a payment gateway.
- Marketing messages need consent and an opt-out.
- Check the local data protection law with a lawyer before launch.
- Camera Form Check: process on the device if possible, do not store video by default, and explain clearly.

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10. Who Does What

Area	Arkesh	Friend
UI design and design system	Yes	Review data needs
Staff web app screens	Yes	
Member PWA	Yes	
Auth, roles, multi-tenant	Login screens	Backend
Members, plans, billing, check-in	Screens, offline check-in	APIs, logic, invoices
Classes, bookings, PT	Screens	APIs, rules
Workouts, progress	Screens, charts	APIs
Messaging and payments	Settings screens	Integrations

Area	Arkesh	Friend
Subscription and admin panel	UI	Backend
AI service (coach, assistant, churn, forecast)	**Build (matches your LLM and FastAPI skills)**	Tools API and data feed
DevOps and backups	Frontend deploy	Server, database, backups

Process: same as the restaurant project. API contract first (OpenAPI), mock server for the frontend, test accounts for each role, weekly call, daily short update.

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11. Phases and Timeline

Rough times for two part-time developers. Adjust after Phase 0.

Phase	What	Time
0. Planning	Finalize this plan, API contract, database, Figma wireframes, show 1-2 real gym staff	1 week
1. Core MVP	Members, plans, billing and dues, QR check-in with offline, staff, basic reports. **Pilot in one real gym**	5-6 weeks
2. Classes, PT, Member app	Timetable, bookings, waitlist, PT sessions, member PWA, WhatsApp reminders	5 weeks
3. Workouts and CRM	Exercise library, plans, logging, progress charts, leads, campaigns, referrals	4 weeks
4. SaaS and Admin	Plans and feature flags, payments, trial, grace period, Super Admin panel	3 weeks
5. AI Heroes	Churn rules then ML, Win-Back Agent, AI Coach, Member Assistant, Crowd forecast, Ask Gym	8-10 weeks
6. Polish and Launch	Speed, security check, user guides, demo gym with fake data, sales page	3-4 weeks

Honest note: this is a 7-9 month job for two people. Finish Phase 1 and test it in a real gym before building the rest. A simple system that works sells better than a big one that is half built.

Tip: do the AI data design in Phase 1 (Section 8.4), even though the AI comes later.

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12. The 3-Minute Demo (Sales Story)

1. A member walks in. The front desk screen shows a green check, photo, and name in 1 second. The live counter goes up.
2. The owner opens the dashboard: **"12 members at risk this week."** One tap on a member: AI drafts a WhatsApp win-back message. Tap Send.
3. A new member answers 6 quick questions in the app. AI creates a workout plan. The trainer approves in one tap. The member sees it.
4. The member asks the assistant: "Book me Zumba tomorrow 6pm." Done, with confirmation.
5. The owner asks: "How is revenue vs last month?" and gets an answer with a chart. Sunday summary arrives on WhatsApp.

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13. Business Notes

- **Pilot gym:** find one friendly gym, give free use for 2-3 months, in exchange for feedback and a testimonial.
- **First selling points:** stops fee leaks (dues reminders), saves front-desk time (2-tap check-in), keeps members (churn alerts).
- **Pricing:** price per month by member count, not per feature only. Check what local gym software costs, and price clearly above "free Excel" but below big foreign tools.
- **Onboarding:** import members from Excel on day 1. If it takes days to start, owners will not switch.
- **Hardware:** a cheap tablet or PC for the front desk, QR on phone first. Door and turnstile integration only after the first paying gyms ask.

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14. Risks

Risk	What to do
Scope is very large	Pilot with Phase 1 only. Build the 5 hero AI features only
AI gives unsafe fitness or diet advice	Trainer approval, library-only exercises, health screen, clear notes, no medical advice
Churn model is not accurate early	Start with rules, show reasons, measure results honestly
Not enough data for ML	Use rules and averages first, say "learning" in the UI
Sinhala and Tamil voice is weak	Test early, text chat first, always show a text confirmation
WhatsApp API setup and cost	Start with SMS and click-to-WhatsApp links, add the API after approval
Members do not use the app	Make QR check-in the reason to open it. Keep the front desk working without the app
Health data leak	Limit access, encrypt, audit log, backups, lawyer review
Internet down at the gym	Offline check-in and offline payment entry with sync later
AI cost grows	Credits and limits per plan, cache answers, small models for simple tasks

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15. First 7 Days

Arkesh:

1. Pick brand name, colours, logo.
2. Figma wireframes: front desk check-in, member profile, add member, member app home.
3. Set up the frontend project using the same structure as the restaurant project.

4. Talk to 2-3 gym owners and a trainer. Ask: "What do you waste time on? What do you lose money on?"

Friend:

1. Choose backend framework and set up repo, Docker, PostgreSQL.
2. Design the multi-tenant database from Section 8.3.
3. Write the OpenAPI contract for auth, members, plans, payments, check-in.
4. Build auth and first endpoints with test accounts.

Together:

- Choose the pilot gym.
- Agree on Phase 1 scope and the weekly call.
- Agree in writing on ownership and income split.

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16. Decisions Needed

1. Is this a separate product from the restaurant system, or one "Ralypto" family of products (shared login, billing, and design system)? **I suggest one family:** you can reuse most of the base (multi-tenant, plans, subscriptions, design system).
2. Target market first: small local gyms (cheap, simple) or bigger gyms and chains (more features, more money)?
3. Do you have a pilot gym in mind?
4. Who builds the AI service? My suggestion: Arkesh.
5. Payments: which gateway and which messaging channel first (SMS or WhatsApp)?